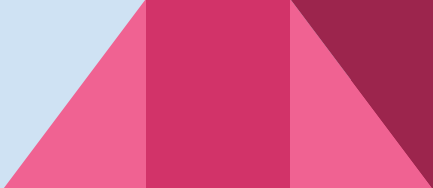




# Lecture 15: Permutations & Combinations

Section 6.3 + 6.5



# Section Summary

- Permutations
- Combinations
- Permutations with Repetitions
- Combinations with Repetitions

# Permutations

**Definition :** A *permutation* of a set of distinct objects is an ordered arrangement of these objects. An ordered arrangement of  $r$  elements of a set is called an  *$r$ -permutation*.

**Example :** Let  $S = \{1,2,3\}$ .

- The ordered arrangement 3,1,2 is a permutation of  $S$ .
- The ordered arrangement 3,2 is a 2-permutation of  $S$ .
- The number of  $r$ -permutations of a set with  $n$  elements is denoted by  $P(n,r)$ .
  - The 2-permutations of  $S = \{1,2,3\}$  are 1,2; 1,3; 2,1; 2,3; 3,1; and 3,2. Hence,  $P(3,2) = 6$ .

# Solving Counting Problems by Counting Permutations (*continued*)

**Example :** Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

**Solution :** The first city is chosen, and the rest are ordered arbitrarily. Hence the orders are:

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

If she wants to find the tour with the shortest path that visits all the cities, she must consider 5040 paths!

# A Formula for the Number of Permutations

**Theorem** : If  $n$  is a positive integer and  $r$  is an integer with  $1 \leq r \leq n$ , then there are

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

$r$ -permutations of a set with  $n$  distinct elements.

**Proof**: Use the **product rule**. The first element can be chosen in  $n$  ways. The second in  $n - 1$  ways, and so on until there are  $(n - (r - 1))$  ways to choose the last element.

- Note that  $P(n, 0) = 1$ , since there is only one way to order zero elements.

**Corollary** : If  $n$  and  $r$  are integers with  $1 \leq r \leq n$ , then

$$P(n, r) = \frac{n!}{(n-r)!}$$

# Solving Counting Problems by Counting Permutations

**Example :** How many ways are there to select a first-prize winner, a second prize winner, and a third-prize winner from 100 different people who have entered a contest?

**Solution :**

$$P(100,3) = 100 \cdot 99 \cdot 98 = 970,200$$

# Solving Counting Problems by Counting Permutations (*continued*)

**Example** : How many permutations of the letters *ABCDEFGH* contain the string *ABC*?

**Solution** : We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*.

$$P(6, 6) = 6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

# Combinations

**Definition** : An *r-combination* of elements of a set is an unordered selection of  $r$  elements from the set. Thus, an *r-combination* is simply a subset of the set with  $r$  elements.

- The number of *r-combinations* of a set with  $n$  distinct elements is denoted by  $C(n, r)$ . The notation  $\binom{n}{r}$  is also used and is called a *binomial coefficient*.

**Example** : Let  $S$  be the set  $\{a, b, c, d\}$ . Then  $\{a, c, d\}$  is a 3-combination from  $S$ . It is the same as  $\{d, c, a\}$  since the order listed does not matter.

- $C(4,2) = 6$  because the 2-combinations of  $\{a, b, c, d\}$  are the six subsets  $\{a, b\}$ ,  $\{a, c\}$ ,  $\{a, d\}$ ,  $\{b, c\}$ ,  $\{b, d\}$ , and  $\{c, d\}$ .



# Combinations

**Theorem** : The number of  $r$ -combinations of a set with  $n$  elements, where  $n \geq r \geq 0$ , equals

$$C(n, r) = \frac{n!}{(n-r)!r!}.$$

**Proof**: By the product rule  $P(n, r) = C(n, r) \cdot P(r, r)$ .  
Therefore,

$$C(n, r) = \frac{P(n, r)}{P(r, r)} = \frac{n!/(n-r)!}{r!/(r-r)!} = \frac{n!}{(n-r)!r!}.$$

# Combinations

**Example** : How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a deck of 52 cards?

**Solution** : Since the order in which the cards are dealt does not matter, the number of five card hands is:

$$\begin{aligned} C(52, 5) &= \frac{52!}{5!47!} \\ &= \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 10 \cdot 49 \cdot 12 = 2,598,960 \end{aligned}$$

- The different ways to select 47 cards from 52 is

$$C(52, 47) = \frac{52!}{47!5!} = C(52, 5) = 2,598,960.$$

*This is a special case of a general result. →*

# Combinations

**Corollary** : Let  $n$  and  $r$  be nonnegative integers with  $r \leq n$ . Then  $C(n, r) = C(n, n - r)$ .

**Proof**: From Theorem 2, it follows that

$$C(n, r) = \frac{n!}{(n-r)!r!}$$

$$\text{and } C(n, n - r) = \frac{n!}{(n-r)![n-(n-r)]!} = \frac{n!}{(n-r)!r!} .$$

Hence,  $C(n, r) = C(n, n - r)$ .

*This result can be proved without using algebraic manipulation. →*

# Combinations

**Example :** How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school.

**Solution :** The number of combinations is

$$C(10, 5) = \frac{10!}{5!5!} = 252.$$

**Example :** A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission?

**Solution :** The number of possible crews is

$$C(30, 6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775 .$$

# Permutations with Repetition

**Theorem** : The number of  $r$ -permutations of a set of  $n$  objects with repetition allowed is  $n^r$ .

**Proof** : There are  $n$  ways to select an element of the set for each of the  $r$  positions in the  $r$ -permutation when repetition is allowed. Hence, by the product rule there are  $n^r$   $r$ -permutations with repetition.



**Example** : How many strings of length  $r$  can be formed from the uppercase letters of the English alphabet?

**Solution** : The number of such strings is  $26^r$ , which is the number of  $r$ -permutations of a set with 26 elements.

# Dealing with Constraints

**Example :** How many bit strings of length 10 contain

a) exactly 4 1s?      b) At most 3 1s?

**a) Solution :** To create a bit string with exactly 4 ones, we need to **choose** 4 positions out of the 10 for the 1s, while the remaining 6 positions will automatically be 0s.

The number of ways to choose 4 positions out of 10 is:  $C(10, 4) = 210$ .

**b) Solution :** We need to consider multiple cases:

**Case 1: Exactly 0 ones:** This corresponds to a string where all 10 bits are 0.

There is only 1 such string.

**Case 2: Exactly 1 one:** Choose 1 position for the 1 out of 10.

The number of ways is:  $C(10, 1)=10$

**Case 3: Exactly 2 ones:** Choose 2 positions for the 1s out of 10.

The number of ways is:  $C(10, 2)=45$

**Case 4: Exactly 3 ones:** Choose 3 positions for the 1s out of 10.

The number of ways is:  $C(10, 3)=120$

**Total:** Now, we use the sum rule for all these values:  $1+10+45+120=176$

# Dealing with Constraints

**Example :** 7 female and 9 male faculties are there in the mathematics department of a school. How many ways are there to select a committee of 5 members from the department if at least 1 female faculty is a must on the committee?

**Solution :** The total number of faculty members is 7 female and 9 male, for a total of  $7+9=16$ . The number of ways to choose 5 members from 16  $= C(16,5) = 4368$ .

Now we need to subtract the number of committees that contain no female, or committees that contain only male. Since there are 9 male, the number of ways to choose 5 members from 9  $= C(9, 5) = 126$ .

Committees with at least 1 female  $= 4368 - 126 = 4242$ .

# Dealing with Constraints

**Example** : How many strings of 6 lowercase letters of the English alphabet contain exactly 2 vowels?

**Solution** : First, we choose 2 positions for the vowels. The number of ways to do this is:  $C(6,2) = 15$ .

Each vowel position can be filled with any of the 5 vowels. Therefore, there are:  $5^2 = 25$  ways to choose the 2 vowels.

The remaining 4 positions must be filled with consonants. Each position can be filled with any of the 21 consonants, so there are:  $21^4 = 194\,481$  ways to choose the 4 consonants.

Finally, we apply product rule to get number of all the strings.  
Total strings =  $15 \times 25 \times 194\,481 = 72\,930\,375$



# Permutations with Indistinguishable Objects

**Example** : How many different strings can be made by reordering the letters of the word *SUCCESS*.

**Solution** : There are seven possible positions for the three Ss, two Cs, one U, and one E.

- The three Ss can be placed in  $C(7,3)$  different ways, leaving four positions free.
- The two Cs can be placed in  $C(4,2)$  different ways, leaving two positions free.
- The U can be placed in  $C(2,1)$  different ways, leaving one position free.
- The E can be placed in  $C(1,1)$  way.

By the product rule, the number of different strings is:

$$C(7, 3)C(4, 2)C(2, 1)C(1, 1) = \frac{7!}{3!4!} \cdot \frac{4!}{2!2!} \cdot \frac{2!}{1!1!} \cdot \frac{1!}{1!0!} = \frac{7!}{3!2!1!1!} = 420.$$